Determining the lower mass limit for central black hole masses detectable in the Virgo cluster by JWST NIRSpec

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Abstract

- The mass function of black holes (BH) at lower masses remains poorly constrained in the local universe due to observational challenges.
- The occupation fraction of BHs within low-mass galaxy hosts like compact stellar systems (CSSs), is essential for competing theories on supermassive BH (SMBH) formation.
- Stellar dynamical modeling have been successful in constraining the masses of central BHs in the cores of CSSs, including nuclear star clusters (NFC), ultra-compact dwarf galaxies (UCDs), and compact elliptical galaxies (cEs).
- We create sets of mock data with ($10^7 < (M_*/M_\odot) < 10^{9.5}$), and construct dynamical models using FORSTAND code.
- **We conclude that a black hole with a mass of a few percent (> 1%) of the total host stellar mass is recoverable.**
- We have obtained JWST+NIRSpec/IFU data cubes to derive spatially resolved stellar kinematics of 18 selected CSSs located in the Virgo Cluster. We will apply our method to determine the black holes mass in these CSSs, which is the first-ever census of CSS BHs in the Virgo galaxy cluster.

CSSs targets and our Mock sample

- We construct N-body spherical distribution functions + central BHs used to represent UCDs, cEs systems in a similar range as our CSSs sample.
- Effective radii, inner slope, velocity anisotropy, and black hole mass fraction varied in mock models (No dark matter halo).

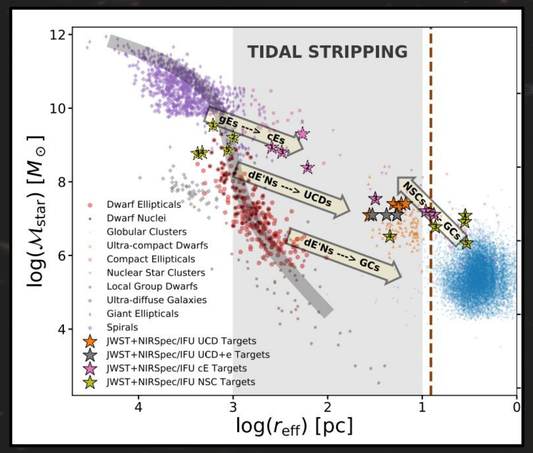

Figure 1: The size-M_* relation for stellar systems in the nearby universe. Stars indicate our targeted sample of CSSs, with dashed outlines indicating cE and NSC host properties. The dashed vertical line illustrates the NIRSpec+IFU spaxel size.

Dynamical Modeling

- Mock samples are projected to the the sky plan and placed at Virgo cluster distance.
- We generate sets of mock kinematic data that match the signal-to-noise ratio (S/N) and field of view (FOV) characteristics as JWST NIRSpec IFU.
- In order to reconstruct the 3d shape from projected images, we use Multi Gaussian Expansion (MGE) methods.
- We use orbit-superposition FORSTAND code to construct dynamical models.

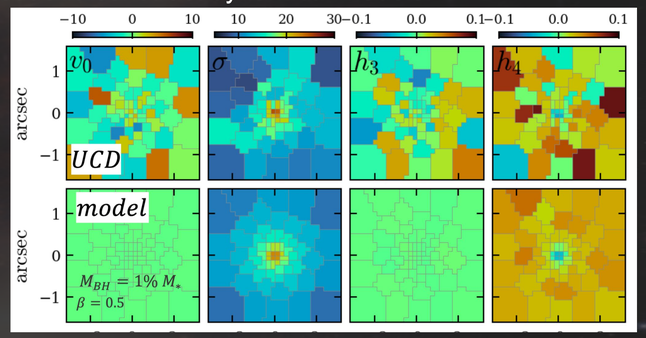

Figure 2: From left to right: line-of-sight velocity, velocity dispersion, and third and fourth order Gauss-Hermite moments. **Top**: Kinematics of a mock UCD-like system. The mock has $M_* = 10^{7.4}$ $M_\odot$ and central anisotropy of $\beta_0 = 0.5$. **Bottom:** The best fitting model.

Identifying a Lower Limit of Recoverable BH Mass

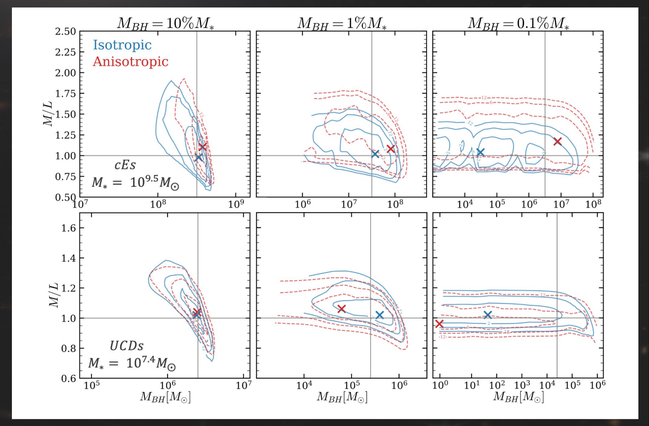

Figure 3: **Top:** The χ^2 plane for cEs with $M_{BH}/M_* = 0.10$, 0.01, and 0.001 respectively. True M/L and M_{BH} values for each mock are denoted with the dashed black cross. For systems with a considerable M_{BH}/M_*, the difference in best-fit values between anisotropic and isotropic realizations is minimal, but this disparity becomes increasingly present for systems with lower M_{BH}/M_*. **Bottom:** The same, but for UCDs.

References

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